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Making Education Accessible and Reliable: Facilitator Based Education Platform

Part IV Project

Project Supervisor : Gill Dobbie





By Moksh Manglani and May Thazin

The Global Education Crisis



44 million additional teachers needed worldwide by 2030 (UNESCO, 2024)

53% of children in low- and middle-income countries can't read a simple text by age 10 (World Bank, 2025)

Under 15% completion rate for unsupervised online learning (Jordan, 2015) — self-directed tech alone isn't the answer

Motivation

I went to school in India and saw this problem firsthand
Skilled teachers kept moving to bigger schools, leaving **low-income students behind**

Two students in the same neighbourhood can have **completely different experiences of the same subject**, just based on which school they landed in

Government schools often **lack the resources** for a proper learning environment

Learning Teachers



Inaccessible

Unreliable



How might we make learning independent of the teachers while being reliable, consistent and easily accessible?







What if **anyone** could teach?

Facilitators!

Facilitators

the person physically in the room with students who Delivers a lesson **without needing subject expertise.**

A Parent, a teaching assistant, a community volunteer, anyone willing to guide a student is a facilitator

Our solution

Admin

Design the lessons



Facilitator

Delivers the lessons

Market research

Platform / Model	Personalisation	Decoupled Delivery	Structured Guidance	Accessibility	Adaptability	Engagement
Formal Schooling (e.g. state schools, K–12)	~	✗	✓	~	~	✓
Homeschooling / SDL (e.g. HSNZ, parent-led)	✓	~	✗	✓	✓	~
MOOCs / Online Self-Paced (e.g. Coursera, Khan Academy)	~	✓	~	✓	~	✗
Blended / Synchronous (e.g. Zoom classes, hybrid)	~	~	✓	~	~	✓
LMS Platforms (e.g. Moodle, Canvas, Google Classroom)	~	~	~	✓	✓	~
Community-Based Models (e.g. volunteer tutoring, co-ops)	~	~	✗	~	~	✓
Our Platform (proposed)	~	✓	✓	✓	✓	~

Splitting the Problem

The Platform & Interviews

The facilitator- and admin-facing system itself, covering the guided lesson-delivery experience

requirements elicitation, interviews, and ethics approval.

Learning Insights

How can less-experienced facilitators be supported in understanding learner progress and providing appropriate learning support?

Demo

StudyBuddy

Quality lessons, ready for anyone to teach.

New facilitators are reviewed by a coordinator before getting access — request access below and you'll hear back by email.

University of Auckland · SOFTENG 700A

StudyBuddy

Request access

Tell us a bit about yourself

Name

Moksh Manglani

Email

Which class/subject are you hoping to help with?

Optional

e.g. Grade 1 Maths

Send request →

Already have an account? Sign in

Over to May

Project # 133

Supporting Facilitator Decision Making

A Decision-Support Framework for Understanding Learner Needs

Part IV Research Project

Project Supervisor : Gill Dobbie



May Thazin

Motivation

Why current approaches leave facilitators with limited insight



Outcomes, not Process

- scores dont show ***how*** a learner got there

Harder for less-experienced facilitators

- pattern spotting takes ***years*** of experience to build



Inconsistent support decisions

- different ***students*** require different ***support***

[1] Van Es, E. A., & Sherin, M. G. (2002). Learning to notice: Scaffolding new teachers' interpretations of classroom interactions. Journal of Technology and Teacher Education.

Can a **software system** convert **learner interaction data** into **evidence-linked recommendations** for **facilitator decision-making**?



Framework Pipeline

Each stage exists to answer one part of the research question — how raw data becomes an evidence-linked recommendation.

1	2	3	4	5
Learning Activity	**Interaction Data**	**Learner Analytics**	**Facilitator Insights**	**Facilitator Decision Making**
Assesses individual students	Captues data **DURING** and **AFTER** the activity	Processes data into **individual** and **class** patterns	Surfaces **key insights** from the data	Facilitator acts on the insights
Keeps every learner's behaviour comparable	Captures **HOW** a learner answered the question	Turns raw signals into **patterns** worth flagging	Every insight is backed by **evidence**	Supports facilitator's judgement – **doesn't replace it**

Proof of Concept & Design Assumptions

Age & Literacy Stage

- Year 3–6 → "learning to read → reading to learn" transition[1]
- Comprehension behaviour: meaningful at this stage

Skill Focus

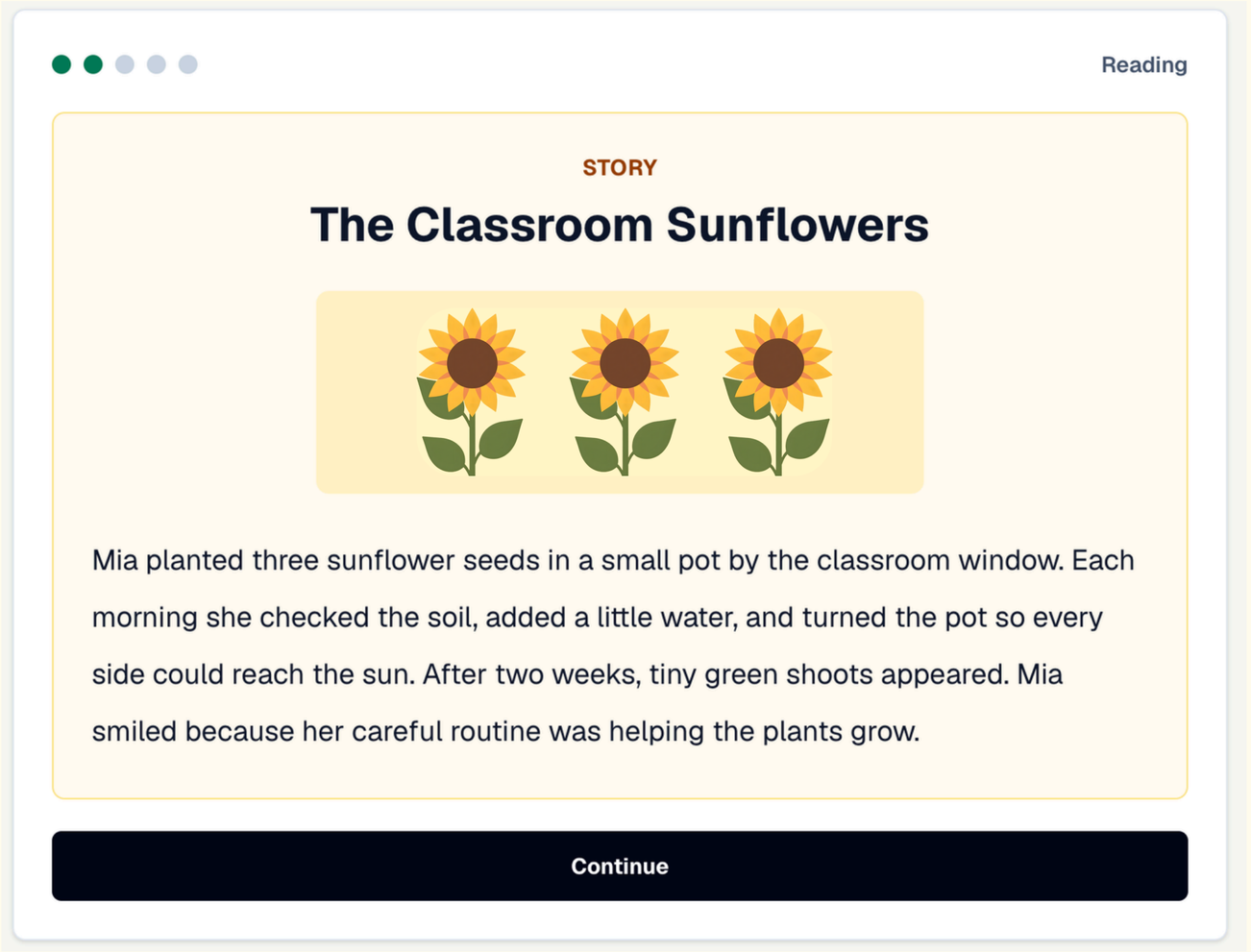
- Decoding: mostly in place by Year 3
- Comprehension: more diagnostic signal

System Advises, Doesn't Diagnose

- Recommendations are flags for a facilitator to interpret
- Never a standalone judgement about a learner

Session Structure

- Short activity before each class
- Behaviour tracked regulary (observing trends)



[1] Chall, J. S. (1983). Stages of Reading Development. New York: McGraw-Hill.

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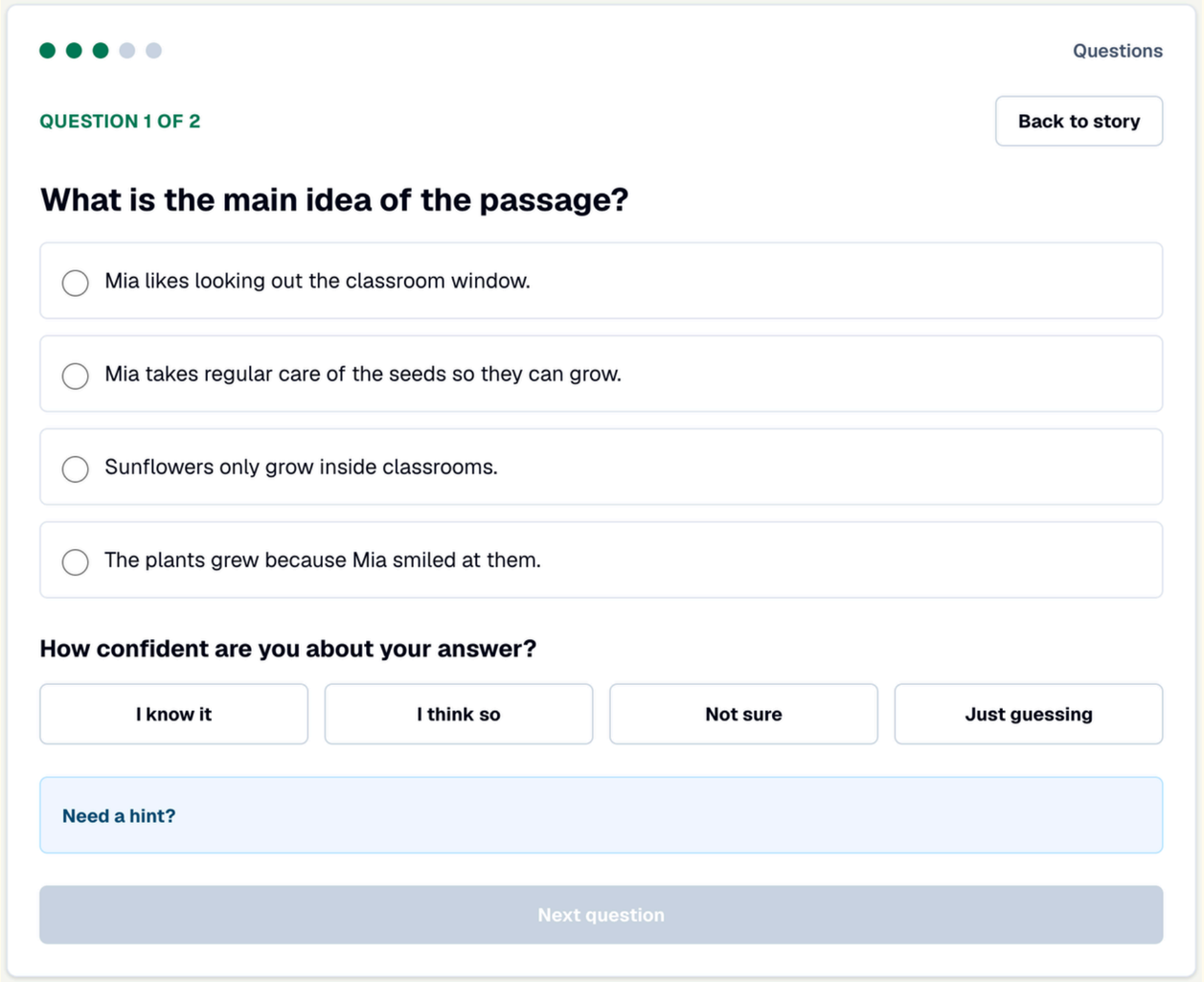
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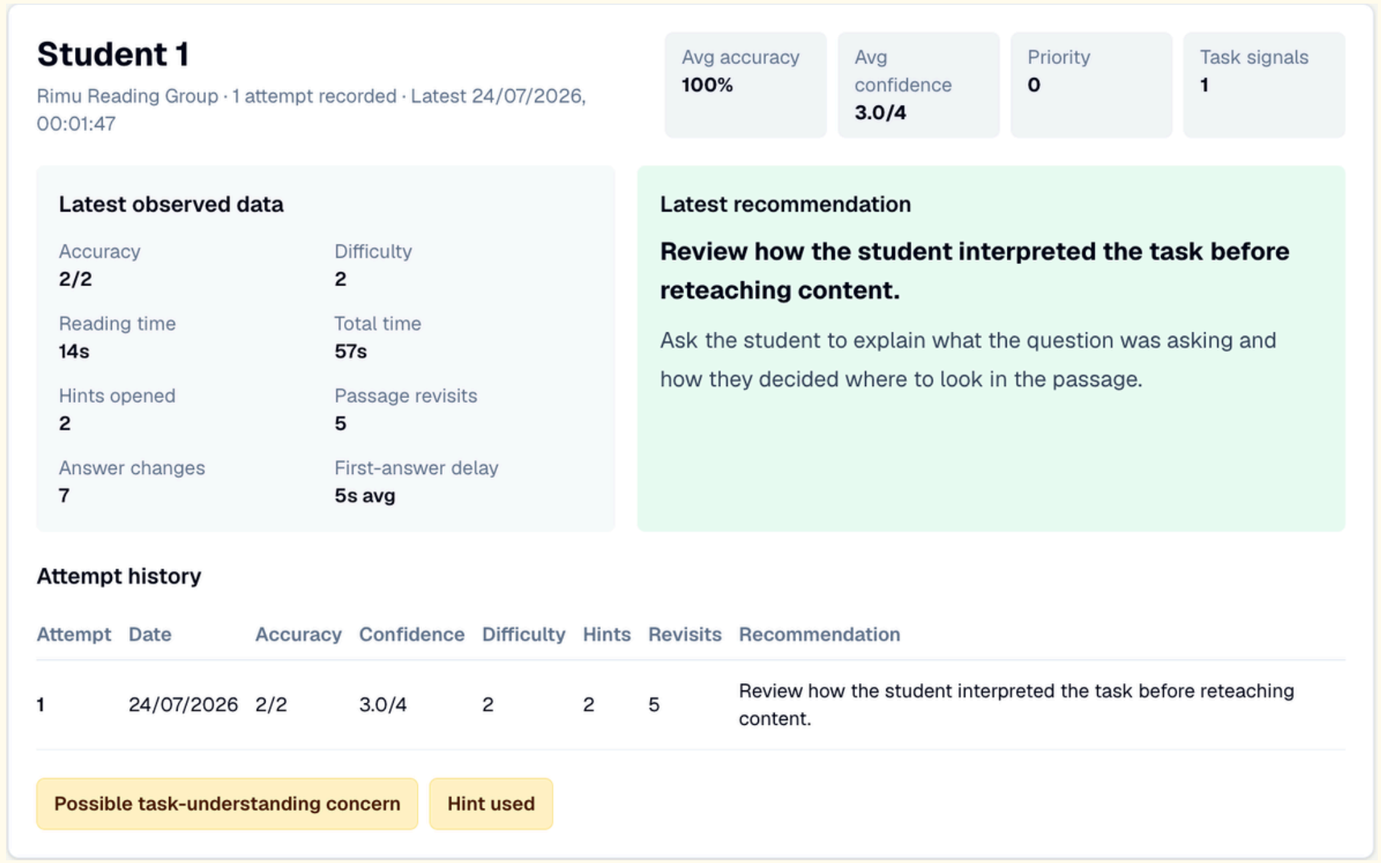
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Proof of Concept & Design Assumptions



Student 1

Rimu Reading Group · 1 attempt recorded · Latest 24/07/2026, 00:01:47

Avg accuracy	Avg confidence	Priority	Task signals
100%	3.0/4	0	1

Latest observed data

Accuracy 2/2	Difficulty 2
Reading time 14s	Total time 57s
Hints opened 2	Passage revisits 5
Answer changes 7	First-answer delay 5s avg

Latest recommendation

Review how the student interpreted the task before reteaching content.

Ask the student to explain what the question was asking and how they decided where to look in the passage.

Attempt history

Attempt	Date	Accuracy	Confidence	Difficulty	Hints	Revisits	Recommendation
1	24/07/2026	2/2	3.0/4	2	2	5	Review how the student interpreted the task before reteaching content.

Possible task-understanding concern | Hint used

Performance

- Accuracy, confidence, and learner achievement.

Interaction Behaviours

- Timing, hints, revisions, and reading behaviours

Learner Insights

- Recommendations, behavioural indicators, and learner history.

Individual and Class Trends

- Progress over time and class-wide learning patterns.

Contributions & Future Work



Contributions

Educational
- Better support for facilitators

Technical
- A general decision-support framework

Research
- Understanding how learner interaction data can inform facilitator decision making

Future Work

Next Steps
- Build out ML comparison
- Rule-based engine → trained models

Beyond the Project
- Modular pipeline — same 5 stages, different activity + rules
- Portable to other subjects and age groups





Thank You!



Q & A



